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Targeting donor XCR1+ and CD11b+ dendritic cells prevents Th1 and Th17-dependent GVHD within the Gastrointestinal Tract.

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Abstract:
Acute graft-versus-host disease (GVHD) of the gastrointestinal (GI) tract is a primary cause of early transplant mortality after bone marrow transplantation (BMT), driven by local antigen presentation and T-cell expansion. We sought to clarify the role of various donor dendritic cell (DC) subsets in this process in order to define therapeutic strategies to prevent gut GVHD. Donor plasmacytoid DC (pDC) were prominent in the ileum of BMT recipients without GVHD but were largely absent in gut during GVHD. In contrast, donor XCR1+ conventional DC (cDC) were dramatically increased in the mesenteric lymph nodes (mLN) of BMT recipients with GVHD. We utilized Xcr1-DTR mice and Clec4c-DTR mice to enable highly efficient XCR1+ cDC versus pDC depletion. Donor XCR1+ cDC but not pDC deletion attenuated lethal GVHD, depleting most cDC presenting alloantigen, which inhibited α4β7 and the expansion of alloantigen-specific donor Th1 cells in the gut. Aldehyde dehydrogenase 1A (ALDH1A) expression by cDC is known to modulate GVHD and we thus examined the effect of a pan-ALDH1 inhibitor, WIN18,446, on this axis. WIN18,446 administration improved survival and these effects were ALDH1A1-specific but mediated by inhibition of CD11b+ rather than XCR1+ cDC, inhibiting alloantigen-specific Th17 cell differentiation in the gut. These findings highlight the limited role of pDC in the induction of gut GVHD and identify differential and dominant roles for XCR1+ and CD11b+ donor cDC in controlling Th1 and Th17 mediated gut aGVHD. The targeting of donor cDC subsets represents an effective strategy to prevent lethal gut GVHD.

Conflict of interest: COI declared - see note

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Targeting donor XCR1$^+$ and CD11b$^+$ dendritic cells prevents Th1 and Th17-dependent GVHD within the Gastrointestinal Tract.

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This paper does not include DNA/RNA sequencing data or high-throughput datasets to be deposited to public repositories. Any additional information required to reanalyze the data reported in this paper is available from the corresponding author (email: mkoyama@fredhutch.org).

Key Points:

- Donor XCR1^{+} cDC present high amounts of alloantigen to induce Th1 differentiation and imprint gut homing integrin expression during gut GVHD.
- Donor CD11b^{+} cDC induce Th17 differentiation during gut GVHD, representing a second pathogenic axis.

Abstract

Acute graft-versus-host disease (GVHD) of the gastrointestinal (GI) tract is a primary cause of early transplant mortality after bone marrow transplantation (BMT), driven by local antigen presentation and T-cell expansion. We sought to clarify the role of various donor dendritic cell (DC) subsets in this process in order to define therapeutic strategies to prevent gut GVHD. Donor plasmacytoid DC (pDC) were prominent in the ileum of BMT recipients without GVHD but were largely absent in gut during GVHD. In contrast, donor XCR1$^+$ conventional DC (cDC) were dramatically increased in the mesenteric lymph nodes (mLN) of BMT recipients with GVHD. We utilized *Xcr1*-DTR mice and *Clec4c*-DTR mice to enable highly efficient XCR1$^+$ cDC versus pDC depletion. Donor XCR1$^+$ cDC but not pDC deletion attenuated lethal GVHD, depleting most cDC presenting alloantigen, which inhibited α4β7 and the expansion of alloantigen-specific donor Th1 cells in the gut. Aldehyde dehydrogenase 1A (ALDH1A) expression by cDC is known to modulate GVHD and we thus examined the effect of a pan-ALDH1 inhibitor, WIN18,446, on this axis. WIN18,446 administration improved survival and these effects were ALDH1A1-specific but mediated by inhibition of CD11b$^+$ rather than XCR1$^+$ cDC, inhibiting alloantigen-specific Th17 cell differentiation in the gut. These findings highlight the limited role of pDC in the induction of gut GVHD and identify differential and dominant roles for XCR1$^+$ and CD11b$^+$ donor cDC in controlling Th1 and Th17 mediated gut aGVHD. The targeting of donor cDC subsets represents an effective strategy to prevent lethal gut GVHD.

Introduction

Allogeneic hematopoietic stem cell/ bone marrow transplantation (hereafter referred to as BMT) remains an important curative therapy for malignant and nonmalignant hematological diseases. GVHD remains the principal therapy limiting toxicity, and early transplant-related mortality is largely dictated by acute GVHD of the GI tract[1,2]. GVHD is initiated by the presentation of alloantigens, either mismatched major histocompatibility complex (MHC) or polymorphic endogenous antigens (minor histocompatibility antigens) within MHC class I or II (MHC-I or MHC-II) on antigen presenting cells (APC) to CD8 or CD4 donor T-cells respectively[3-6]. Gut GVHD is a highly inflammatory MHC-II dominant process in which CD4 T-cell derived cytokines and cytolysis are pivotal.[3,7] With this in mind, understanding the processes by which MHC-II dependent alloantigen presentation occurs in the GI tract is critical for the design of rational therapeutic strategies to interrupt this process.

Recipient hematopoietic and non-hematopoietic APC are critical to the initiation of gut GVHD,[8-10] while the maintenance and amplification of disease is mediated by donor APC.[11] Within donor professional APC, dendritic cells (DC) are the principal population that present host alloantigen within MHC-II[11-13] and MHC-I.[14] DC include a number of subsets by phenotype and function. Conventional DC include $CD8^+CD103^+$ or $XCR1^+$ (cDC1) subsets that specialize in cross-presentation of exogenous antigen within MHC-I and $CD11b^+$ (cDC2) subsets that are known to be highly efficient at presentation of exogenous antigens within MHC-II[15]. Plasmacytoid DC (pDC) represent a distinct subset that produce high levels of type I interferons in response to pathogen-derived nucleic acids and are important in viral immunity[16]. Critically, the standard immunological paradigms regarding antigen presentation by various DC subsets are highly perturbed after stem cell transplantation, reflecting a biological process that has no evolutionary peer since alloantigen is present in vast amounts and indefinitely.

Thus, the examination of alloantigen presentation in the gut after BMT requires analysis without the bias of prior immunology dogma acquired at steady state. While it has been established that CD103$^+$ cDC present large amounts of alloantigen in the colon during GVHD, cause–and–effect studies of this effect in GVHD that include cell subset deletion have not been possible to date. In terms of the role of pDC in gut GVHD, data to date is ambiguous. Donor pDC have been shown to ameliorate GVHD and T-cell function[17], whereas ICOSL-expressing pDC were associated with GVHD lethality and Th17 differentiation[18]. To add to this complexity, the development of pDC during GVHD is also highly perturbed[17,19].

We thus sought to analyze the role of donor APC subsets in acute GVHD with new tools that permit specific DC subset depletion. XCR1, a chemokine receptor, has been shown to be expressed exclusively by CD8$^+$ DC and CD103$^+$CD11b^{neg} DCs[20,21]. The *Xcr1*-diphtheria toxin receptor (DTR) venus mouse has a fusion protein containing a DTR and a Venus reporter protein knocked into *Xcr1* locus, permits us to deplete donor XCR1$^+$CD103$^+$CD11b^{neg} DC via diphtheria toxin (DT) administration[22]. C-type lectin domain family 4 member C (CLEC4C), known also as blood dendritic cell antigen 2 (BDCA2), is encoded in humans but not in mice, however, the pDC-specific transcription factor, E2-2, interacts with elements of the CLEC4C promoter in mice, resulting in feasible pDC-specific depletion in *Clec4c*-DTR mice following DT administration[23]. In this study, we demonstrate that selective depletion of donor XCR1$^+$ cDC but not pDC mitigates the severity of GVHD by suppressing Th1 differentiation and the imprinting of gut-homing receptor on donor T-cells. In addition, in the process of exploring pharmacological approaches to target cDC, we noted that a broad inhibitor of aldehyde dehydrogenase enzymes, WIN18,446, suppressed donor CD11b$^+$ cDC expansion, also resulting in the attenuation of GVHD via reduction of Th17 differentiation. These studies highlight differential roles of donor cDC subsets in generating pathogenic donor T-cell differentiation after BMT and provide rational approaches for therapeutic intervention.

Methods

Mice. Female, 8- to 12-week-old C57BL/6 (B6), *Clec4c*-DTR (C57BL/6-Tg(CLEC4C-HBEGF)956Cln/J, stock number 014176), BALB/c and B6D2F1 mice were purchased from Jackson Laboratory (Bar Harbor, Maine). *Aldh1a1*$^{-/-}$ (B6.129-*Aldh1a1*^{tm1Gdu}/J, stock number 012247) mice were cryo-recovered from the Jackson Laboratory and backcrossed to C57BL/6 for >10 generations. BALB/c mice were purchased from Charles River Laboratory (Wilmington, MA). B6.*Xcr1*-Venus, B6.*Xcr1*-DTR Venus, MHC-II knockout, CD11c-GCDL and B6.TEa mice have been previously described [11,22,24,25,26] and were bred in-house. We utilized heterozygous B6.*Xcr1*-Venus, B6.*Xcr1*-DTR Venus or Clec4c-DTR mice to avoid targeted gene deficiency in homozygous mice. [27] B6.TEa x *Rag1*$^{-/-}$ mice and B6.TEa x luciferase mice were generated by back crossing B6.TEa mice with B6.*Rag1*$^{-/-}$ or B6.luciferase mice, respectively. Mice were housed in sterilized cages and received sterile water and normal chow unless the individual indications were noted for the diet treatment. All animal procedures were performed in accordance with the protocols approved by Institutional Animal Care and Use Committee guidelines (IACUC 51055).

Bone marrow transplantation (BMT). MHC-mismatched BMT was performed as described previously[9,11]. Briefly, recipient B6D2F1 (H-2$^{b/d}$) or BALB/c (H-2^{d}) mice were received 1100cGy or 900cGy total body irradiation (^{137}Cs source) on day -1, respectively and transplanted with BM with or without T-cells from indicated B6-background (H-2^{b}) donor mice on day 0. T-cell-depleted BM (TCD-BM) was prepared by anti-mouse CD4, CD8 and Thy1.2 antibodies and complement (CEDARLANE, Burlington, Canada). T-cells from B6 spleen and peripheral lymph node were purified using anti-mouse GR1, CD11b, B220, CD19 antibodies and BioMag Goat anti-Rat IgG (QIAGEN, Hilden, Germany). TEa CD4 T cells from B6.TEa mice were isolated using CD4 isolation kit (Miltenyi Biotec, NRW, Germany) or were FACS sorted (Purity> 98%; Vα2^{+}Vβ6^{+}). In

B6→B6D2F1 transplants, irradiated B6D2F1 mice received 5 or 10 × 10^6 TCD-BM from indicated B6-background mice with 2 × 10^6 B6.WT T-cells on day 0. The recipients of TCD graft or B6 recipients (syngeneic) served as non-GVHD controls. In B6→BALB/c transplants, irradiated BALB/c mice were transplanted with 10 × 10^6 TCD-BM from indicated B6-background mice with 0.2 × 10^6 B6.WT or TEa T-cells (doses detailed in the figure legend). To induce polyclonal CD4 T-cell-mediated GVHD, whilst minimizing the influence of host APC, lethally irradiated B6D2F1 mice were transplanted with 10 × 10^6 B6-background wild-type or transgenic (CD45.2$^+$) TCD-BM on day 0 followed by 2 × 10^6 B6 (CD45.1$^+$) CD4$^+$ T-cells on day 3 when host APC are largely eliminated [28]. For reporter assays, BALB/c (I-A^d I-E^d) recipients received B6-background (I-A^b I-E^{neg}) BM and T-cells on day 0, then 2-3 ×10^6 luciferase-expressing TEa T-cells (reacting to the I-E-derived peptide presented in I-A^b) on day 12, when donor APC are fully reconstituted [11,12]. Recipients were monitored daily and clinical scores measure after transplantation. Recipient mice were euthanized when their clinical score reached ≥6 or weight loss was over 30% at any time [29]. For WIN18,446 treatment, recipient mice were treated with WIN18,446 or control diet from day -7 to endpoint of experiments. WIN18,446 diet (AIN93M containing 2 mg WIN 18,446/g) and its control diet (AIN93M) were produced by TestDiet (Richmond, IN). [30]

Diphtheria toxin injection. For survival experiments, 250 ng diphtheria toxin (DT) for donor B6 and B6.*Xcr1*-DTR Venus mice and 100 ng DT for donor B6.*Clec4c*-DTR mice were intraperitoneally injected on day -2 and day -1. All recipient BALB/c mice were treated with 100 ng DT every other day from day 3 to day 21 after transplant. For TEa adoptive transfer experiments, 160 ng DT was injected every other day from day 8 to the experimental endpoint.

Antibodies. Antibodies used for flow cytometry were purchased from BD (Franklin Lakes, NJ), BioLegend (San Diego, CA), and eBiosience (San Diego, CA). Antibody details are described in Supplemental Table 1.

Cell preparation for flow cytometry. Recipient mice were euthanized and their spleen, mesenteric lymph node and gut were harvested. Spleen samples underwent red blood cell lysis prior to surface staining. Gut digestion was performed as described previously[9,11]. Small intestine and colon were cut longitudinally, washed with PBS 3 times, and cut into 3-5 mm pieces. The pieces were incubated under agitation in Calcium/Magnesium-free HBSS containing 10mM HEPES, 5mM EDTA, and 5% FCS for 20 minutes at 37°C. Cells were isolated by passing through a 100um cell strainer. The cell suspension was kept as the intraepithelial leukocyte fraction (epithelium fraction). The remaining tissue was processed with a mouse Lamina Propria Dissociation kit and gentleMACS Dissociator (Miltenyi Biotec) according to the manufacturer's protocol. After incubation, the cell suspension was passed through a 100um cell strainer to obtain the lamina propria leukocyte fraction (lamina propria fraction). Fc blocking was performed using purified rat anti-mouse CD16/32 antibody (2.4G2, BD). After Fc blocking, cells were stained with anti-E52-68 peptide/I-A^b complex (YAe) antibody followed by antibodies for other surface markers. For intracellular cytokine staining, cells were stimulated for 4 hours at 37°C with phorbol myristate acetate (PMA; 5 mg/mL) and ionomycin (50 mg/mL) (Sigma-Aldrich, St. Louis, MO) in the presence of Brefeldin A (BioLegend). After stimulation, cells were surface stained, then fixed and permeabilized using a Foxp3 transcription factor staining kit (eBioscience) before intracellular staining. All samples were acquired on a BD FACSymphony A3 Cell Analyzer (BD Biosciences) and analyzed using FlowJo software (TreeStar, Ashland, OR).

Bioluminescence imaging. See supplemental data.

Immunofluorescence microscopy. See supplemental data.

Statistics. Data were analyzed using GraphPad Prism (ver. 10) (La Jolla, CA). All hypothesis tests were two-sided unless otherwise specified, with statistical significance defined as $p<0.05$. Survival curves were estimated using Kaplan-Meier method and compared using the log-rank test. If the data was normally distributed, the Brown-Forsythe ANOVA test and Welch's t-test were used to compare groups. P-values were adjusted to account for multiple pairwise comparisons using Dunnett's procedure. For data that was not normally distributed, the Kruskal-Wallis test and the Mann-Whitney U test were used to compared groups. P-values were adjusted to account for multiple pairwise comparisons using Dunn's multiple comparison procedure. Data were presented as mean ± standard error of the mean (SEM).

Results

Donor pDC and cDC differentially expand within the GI tract.

In order to understand the anatomical distribution of DC in the GI tract and lymphoid organs after BMT (B6 →B6D2F1), we undertook flow cytometry analysis of recipients transplanted with B6.XCR1 reporter (*Xcr1*-venus$^{+/-}$) donor grafts that facilitate detection of the CD8^{+}/CD103^{+}CD11b^{neg} cDC subset.[22] On day 14 after BMT, donor SiglecH^{+} pDC (CD45^{+}H2D^{d-} ^{neg}CD3^{neg}CD19^{neg} Ly6G^{neg}CD64^{neg}CD11b^{neg}B220^{+}PDCA1^{+}SiglecH^{+}) were prominent in the ileal but not colonic epithelium of non-GVHD (T-cell deplete) compared to GVHD (T-cell replete) recipients (**Figure 1A, Supplemental Figure 1A-B**). CD103^{+}CD11b^{neg} (termed single CD103^{+} [shown as sCD103^{+}]) cDC (CD45^{+}H2D^{d-} ^{neg}CD3^{neg}CD19^{neg}Ly6G^{neg}CD64^{neg}B220^{neg}CD11c^{+}IAIE^{+}CD103^{+}CD11b^{neg}) were decreased in the lamina propria of ileum and colon of GVHD recipients relative to non-GVHD control (**Figure 1A**). Consistent with their known migration from the gut into the draining mLN during GVHD,[11] donor CD103^{+}CD11b^{neg} cDC were significantly increased in the mLN of recipients with GVHD (**Figure 1A, Supplemental Figure 1A**). To quantify alloantigen presentation by donor APC, we utilized the YAe antibody which binds to the host-derived Ea (I-E^{d}) peptide loaded on donor I-A^{b}.[11] YAe^{+} cells were enriched in the CD103^{+}CD11b^{neg} cDC fraction. BMT recipients with GVHD had significantly increased frequencies of YAe^{+} CD103^{+}CD11b^{neg} cDC in the mLN and, to a lesser extent, colonic lamina propria compared to non-GVHD mice. In contrast, pDC were YAe negative in all tissues, consistent with minimal alloantigen presentation by this subset (**Figure 1B**). YAe expression (measuring alloantigen presentation) by donor APC subsets during GVHD was most dramatic within cDC in the mLN (Figure 1A-B), with donor macrophages representing a minor subset, consistent with our previous work demonstrating they are not a major alloantigen presenting cell-type[11]. Of note, we confirmed that, with the exception of non-GVHD spleen, donor *Xcr1*-venus expression after transplant was restricted to CD103^{+}CD11b^{neg} cDC (**Figure 1C**).

Donor cDC rather than pDC amplify acute GVHD.

We next examined the influence of donor CD103^{+}CD11b^{neg} cDC and pDC on GVHD outcome. We utilized *Xcr1*-DTR Venus [22] (hereafter referred to as *Xcr1*-DTR) mice and *Clec4c*-DTR [23] mice in which the DTR is expressed under the promoter of XCR1 or Clec4c respectively, to permit isolated donor-derived CD103^{+}CD11b^{neg} cDC and pDC depletion by DT administration (**Supplemental Figure 2A – C**). To assess GVHD induced by donor APC-mediated antigen presentation in isolation, lethally irradiated BALB/c mice (H-2^{d}) were transplanted with TCD-BM from B6 wild type (B6.WT), *Xcr1*-DTR or *Clec4c*-DTR mice (all H-2^{b}) with CD4^{+} *Rag1*$^{-/-}$ TEa transgenic T-cells reactive to the BALB/c-derived I-E^{d} peptide presented on donor B6 MHC-II (I-A^{b}), resulting in the induction of donor APC-mediated GVHD in isolation. Recipients of TCD BM alone from B6.WT mice served as non-GVHD controls. DT was administered to donor mice on day -2 and day -1 and on alternate days to recipients from day +3 to day +21. Donor XCR1^{+} cDC depletion but not pDC depletion attenuated GVHD lethality compared to non-depleted controls (**Figure 2A**), consistent with XCR1^{+} cDC being a major determinant of acute GVHD pathology. We confirmed that the depletion of donor XCR1^{+} cDC also mitigated lethal GVHD in a polyclonal CD4^{+} T-cell-mediated system **(Figure 2B).** To quantify XCR1^{+} cDC and pDC interaction with T-cells, we measured CD4^{+} or CD8^{+} cell numbers within a 15 μm distance of XCR1^{+} cDC or SiglecH^{+} pDC on day 14 after transplant. Given cell size (< 10 μm for T-cells and <15 μm for DCs), we posited that a proximity of 15 μm likely reflects cell-to-cell contact and putative antigen presentation of DC to T-cells. We normalized the T-cell counts to 10,000 cells per field. CD4^{+} and CD8^{+} T-cells in close proximity to donor XCR1^{+} cDC were increased in the colon and mLN during GVHD, relative to SiglecH^{+} pDC (**Figure 2C–D**). Of note, vicinal CD4^{+} and CD8^{+} T-cell accumulation to donor XCR1^{+} cDC were prominent in the mLN after allogeneic BMT compared to those in the colon (**Figure 2C–D**). In sum, CD4^{+} and CD8^{+} T-cells accumulate and are in close proximity to donor XCR1^{+} cDC but not to SiglecH^{+} pDC in the mLN during GVHD.

Donor XCR1$^+$ cDC expand alloreactive Th1 cells and induce integrin α4β7 expression during GVHD.

Next, we examined donor XCR1$^+$ cDC-driven alloreactive T-cell expansion among various organs using our previously established model in which the expansion of luciferase reporter CD4$^+$ TCR transgenic cells (TEa^{luc+}) reactive to donor APC reactive presenting alloantigen are quantified 3-5 days following transfer at a time when donor DC are fully reconstituted.[11] Lethally irradiated BALB/c (I-A$^{b\ neg}$) mice were transplanted with TCD-BM from B6.WT (I-A^{b+}), *Xcr1*-DTR (I-A^{b+}), or MHC-II deficient (I-A$^{b\ neg}$) B6 mice along with 0.2 x 10^6 T-cells from B6.WT mice. DT (160 ng/ dose) was administered to all recipients from day 8 after transplant and recipients of MHC-II deficient BM served as a negative control for TEa responses. TEa^{luc+} were transferred into BMT recipients on day 12 when donor APC are fully reconstituted.[12] TEa expansion was assessed by bioluminescence 5 days later (day 12+5) (**Figure 3A**). While whole body bioluminescence signal was increased in donor MHC-II intact mice but was not significantly different in between XCR1$^+$ cDC-depleted and non-depleted GVHD recipients (**Figure 3B**), TEa expansion was decreased in the GI tract and mLN in the absence of XCR1$^+$ cDC (**Figure 3C**). In contrast, donor pDC-depletion of recipients (DT-treated *Clec4c*-DTR BM recipients) did not mitigate TEa expansion in the GI tract (**Supplemental Figure 3A – B**). We verified that DT treatment depleted the donor XCR1$^+$ cDC subset in the mLN, colon and spleen (**Figure 4A**) but did not alter the total cDC numbers (**Supplemental Figure 4A**). Thus, alloreactive TEa T-cell expansion primarily reflected the function of donor XCR1$^+$ cDC. Consistent with this, donor XCR1$^+$ cDC-depletion resulted in a reduction of YAe$^+$ alloantigen presenting DC in the mLN during GVHD (**Figure 4B**). At a qualitative level, alloantigen presentation (i.e. YAe fluorescence intensity) by the CD103$^+$CD11b^{neg} cDC subset was dramatically higher than in any other cDC subset (CD103$^+$CD11b$^+$ or CD103^{neg}CD11b$^+$), which were not affected by XCR1 depletion (**Figure 4C**). Thus, donor XCR1$^+$ cDC depletion eliminated the major alloantigen presenting cell subset from the mLN. Since donor DC in the mLN have been shown to present alloantigen and dictate CD4$^+$ T-cells to migrate into

the gut[11], we quantified the expression of integrin α4β7, a gut-homing receptor, on TEa T-cells in the mLN. The depletion of XCR1$^+$ cDC attenuated integrin α4β7 expression and the number of α4β7$^+$ TEa cells in the mLN (**Figure 4D – E**). Given that mLN cDC, particularly CD103$^+$CD11b^{neg} cDC produce high amounts of IL-12[11], XCR1$^+$ cDC depletion may affect T-cell differentiation. Indeed, the generation of interferon-γ (IFN-γ) secreting TEa T-cells was also significantly reduced by donor XCR1$^+$ cDC depletion (**Figure 4F**). Thus, donor XCR1$^+$CD103$^+$CD11b^{neg} cDC are pivotal in the generation of alloreactive Th1 differentiation and expansion in the GI tract during GVHD.

WIN 18,446 inhibits donor CD11b$^+$ cDC and allo-antigen specific Th17 differentiation.

We next analyzed pharmacological strategies to inhibit cDC function during GVHD, using WIN18,446, a broad inhibitor of aldehyde dehydrogenase enzymes of the 1A family (ALDH1A) that has been administered in preclinical and clinical settings. [31,32] ALDH1A which includes isoforms, ALDH1A1, ALDH1A2 and ALDH1A3, is required to catalyze the oxidation of retinal to retinoic acid (RA), and *Aldh1a2* gene deletion in donor and host DC has previously been shown to attenuate GVHD.[33] The oral administration of WIN18,446 improved survival in a system where GVHD lethality is solely reliant on MHC-II dependent donor APC function (**Figure 5A**). WIN18,446 treatment diminished donor DC (luciferase-expressing CD11c$^+$) expansion including within the GI tract (**Figure 5B**). This resulted in reduced alloreactive donor T-cell expansion in the GI tract (**Figure 5C**). Unexpectedly, the CD11b$^+$ cDC subsets (CD11b$^+$CD103$^+$ and CD11b$^+$CD103^{neg}) but not CD103$^+$CD11b^{neg} cDC were impacted by WIN18,446 treatment (**Figure 5D**). WIN18,446 diminished alloreactive T-cell expansion systemically at later time points after BMT (**Figure 5E**), in association with suppression of Th17 differentiation in the mLN (**Figure 5F**). The combination of XCR1$^+$ DC depletion and WIN18,446 treatment reduced the absolute number of both CD103$^+$ and total cDC and relevant T-cell differentiation in the mLN (**Figure 5G**).

We investigated whether the effect of WIN18,446 after transplantation was mediated by ALDH1A1, one of the major isoforms of ALDH1A. We compared alloantigen presentation, quantified by TEa expansion, in recipients of B6.WT and B6.*Aldh1a1*$^{-/-}$ BM with or without WIN18,446 treatment. Of note, the *Aldh1a1* gene remained intact in the TEa reporter T-cells. The effects of WIN18,446 on TEa expansion and Th17 differentiation were abolished in the recipients of *Aldh1a1*$^{-/-}$ BM (**Figure 6A – B**). Conversely, the effects of WIN18,446 on α4β7 expression were independent of ALDH1A1 (**Figure 6C**). Thus, WIN18,446 suppresses alloreactive Th17 differentiation and expansion via ALDH1A1 expression in CD11b^{+} cDC. We demonstrate the reduction of CD11b^{+} cDC expansion, Th17 differentiation and α4β7 integrin expression on CD4 T-cells by WIN 18,446 treatment after BMT in a model where GVHD was induced by polyclonal CD4^{+} T-cells (**Supplemental Figure 5A**). We confirmed that the reduction in CD11b^{+} cDC by WIN 18,446 treatment was in both the CD11b^{+}CD103^{+} and CD11b^{+}CD103^{neg} fractions that exist within lymph nodes (**Supplemental Figure 5B**).

Discussion

Alloantigen presentation represents the critical initiating event that drives donor T-cell responses and the subsequent manifestations of GVHD. While host APC potently initiate acute GVHD[8,10,34,35] and are pivotal for maximal graft-versus-leukemia (GVL) effects[6], professional host APC can also constrain donor T-cell activation via the induction of activation-induced cell death[8,36,37]. Conversely, donor hematopoietic APC are not required for GVL, but maintain and amplify acute GVHD, and thus represent an appealing target for the prevention and treatment of acute GVHD[4,11]. Clinical studies to date demonstrate that GVHD suppresses cDC and increase pDC numbers in peripheral blood.[18,38,39] We now show that donor cDC numbers are also reduced in the gut during early preclinical GVHD, with accumulation in mLN, where T cell priming by donor DC occurs.[11] Appropriate correlative clinical analysis of these findings will therefore require tissue analysis taken at postmortem that permits access to large quantities of tissue (rather than diagnostic pinch biopsies), including sampling of mesenteric lymph nodes where DC – T cell interactions occur. To date no study has analyzed donor DC subsets in gut tissues from patients with GVHD. Importantly, excellent recent studies have analyzed clinical GVHD gut biopsies but lack the critical mesenteric lymph node tissue and have used methods that allow analysis of approximately 40 proteins, which lack resolution to determine DC versus macrophage subsets.[40]

In order to design rational therapeutics targeting donor APC, it is essential to identify which subsets enhance GVHD. However, GVHD severity in BMT models where donor T-cells are transferred immediately after TBI are predominantly determined by host APC that confound the analysis of effects mediated by donor DC. To overcome this, we leveraged a GVHD system in which donor $CD4^+$ T-cells (TEa T-cells) can only be activated by reconstituting donor APC presenting alloantigen in isolation. We thus transferred TEa TCR Tg T-cells on day 12 when donor APC are fully reconstituted and assessed the roles of donor alloantigen presentation on

alloantigen specific T-cell expansion and differentiation thereafter. Here we confirm that both CD103$^+$ and CD11b$^+$ cDC, but not pDC, are critical for the induction of lethal acute GVHD by differentially driving pathogenic Th1 and Th17 differentiation.

While our previous work demonstrated that donor CD103$^+$CD11b^{neg} cDC play a primary role in alloantigen presentation in the mLN[11], recent studies suggest that donor pDC can either promote or suppress GVHD[17,18]. However, to date, studies have lacked cell type-specific depletion interventions to provide direct evidence for pDC involvement in the pathogenesis of GVHD. By utilizing extensive tissue characterization with *Xcr1*-DTR and Clec4c-DTR mouse strains, we have been able to directly assess the impact of donor CD103$^+$CD11b^{neg} cDC and pDC in GVHD pathogenesis. These studies clearly show that donor CD103$^+$CD11b^{neg} cDC are the definitive antigen presenting population in the gut that promotes GVHD. In contrast, donor pDC depletion did not significantly modulate T-cell expansion or acute GVHD. This is consistent with the expansion of these cells after BMT explicitly in the ileal epithelium, and in the absence of GVHD. Their role in this setting remains to be elucidated.

The administration of WIN18,446, a pharmacological pan-ALDH1A inhibitor that blocks RA biosynthesis[41] acted on CD11b$^+$ cDC expansion. RA is the active metabolite of vitamin A and synthesized by ALDH1As, which are highly expressed by DC and macrophages, epithelial and stromal cells in a tissue-specific manner[41,42]. Given that prior studies demonstrated that 1) GVHD induces both ALDH1A1 and ALDH1A2 expression in both nonhematopoietic and hematopoietic murine cells[43] and 2) both the accelerated RA degradation (via CYP26A1 induction) in recipient or donor hematopoietic components, and the absence of ALDH1A2 in host or donor DC mitigates GVHD we hypothesized that WIN18,446 may impact cDC homeostasis via inhibition of ALDH1A1 and ALDH1A2 isoforms. We note that to date, ALDH1A1 function has not been examined in hematopoietic cells after BMT.[33] We noted that WIN18,446 mitigated donor APC-mediated GVHD lethality unexpectedly via donor CD11b$^+$ cDC rather than CD103$^+$CD11b^{neg} cDC, with subsequent

attenuation of Th17 differentiation. These effects of WIN18,446 were abrogated when the BM graft was ALDH1A1 deficient, suggesting the decrease of Th17 differentiation is an indirect effect on T-cells from the absence of ALDH1A1 in donor APC. These results suggest that not only ALDH1A2 but ALDH1A1 isoform modulate donor APC function, however, the relative importance of these isoforms has not yet been established. These cellular sequential effects are consistent with the findings by previous studies; $CD11b^{+}$ DC, particularly $CD103^{+}CD11b^{+}$ cDC, rather than $CD103^{+}CD11b^{neg}$ DC or macrophages produce the largest amounts of IL-6, which promotes Th17 differentiation and exacerbates lethal acute GVHD. [44-47] Donor T-cells deficient in IL-17A or *RORC* encoding RORγt, the essential transcription factor for Th17 differentiation, mitigate acute GVHD [48-50] and patients with acute GVHD or active chronic GVHD have increased Th17 cells in blood[51] and tissue[52], consistent with the contribution of Th17 cells to GVHD. The indirect effect of WIN18,446 on T-cell differentiation contrasted with previously reported direct effects of RA inhibition on donor T-cells that skewed differentiation from Th1 differentiation to Th2 and Th17 differentiation during GVHD [43]. Unlike the effects on T-cell differentiation, the expression of α4β7 integrin by donor T-cells is likely to be directly modulated by RA signaling[53-55]. Importantly, the WIN18,446 compound has been widely tested in clinical studies and here we have demonstrated that the compound is highly efficient at inhibiting donor APC function and protecting from lethal GVHD. The compound would thus be highly testable in translational studies as an adjunct to prevent GVHD.

In conclusion, our findings highlight the limited role of pDC in the pathophysiology of gut GVHD and identify differential and dominant roles for $XCR1^{+}$ $CD103^{+}CD11b^{neg}$ and $CD11b^{+}$ donor cDC in controlling Th1 and Th17 mediated gut aGVHD. Strategies to control gut GVHD by pharmacological intervention should ideally target both cDC subsets.

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Authorship Contributions

ST, TI, and MK conducted experiments and analyzed the data. ST helped to write the manuscript. KSE, SRWL, TS, EN, NSN, TJ, SAM, ACY and PZ helped perform experiments. MH, JP and JKA helped design experiments and reviewed the manuscript. GRH and MK conceived and designed the studies and wrote the manuscript. All authors reviewed, critically edited, and approved the final manuscript for submission.

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Figure legends

Figure 1. Donor XCR1$^+$ CD103$^+$ cDC expand and present alloantigen during GVHD.

Recipient B6D2F1 mice received 1100cGy TBI and bone marrow from B6. *Xcr1*-venus$^{+/-}$ mice with or without 2 × 10^6 T-cells from B6.WT mice. The ileum and colon (epithelium and lamina propria), mLN and spleen analyzed on day 14 (*n* = 6 – 8 / group from two independent experiments). (**A**) Representative flow plots from mLN and the absolute number of donor cells on each subset. (**B**) Representative flow plots from mLN and the frequency of YAe$^+$ donor cells in each subset. (**C**) Frequency of *Xcr1*-venus$^+$ in each subset. sCD103$^+$ DC, dp DC and sCD11b$^+$ DC represent CD103$^+$CD11b^{neg} DC, CD103$^+$CD11b$^+$ DC, and CD103^{neg}CD11b$^+$ DC, respectively. Mac = macrophages. Individual populations of GVHD and non-GVHD group were compared using Mann-Whitney test, dictated by normality testing of the data (mean ± SEM). **P* <0.05, ***P* <0.01, ****P* <0.001.

Figure 2. Depletion of XCR1$^+$ cDC ameliorates GVHD.

(**A**) Recipient BALB/c mice received 900cGy TBI and were transplanted with 10 × 10^6 TCD-BM from DT-treated B6. *Xcr1*-DTR, B6. *Clec4c*-DTR, or B6.WT mice with 1.5 × 10^6 Rag1$^{-/-}$ TEa cells. Recipients of B6 TCD BM alone served as non-GVHD controls. Recipient mice were treated with DT every other day from day 3 to 21 after BMT. GVHD survival and clinical score were monitored after transplant (*n* = 6 – 10 / group from two independent experiments). (**B**) lethally irradiated B6D2F1 mice were transplanted with 10 × 10^6 B6 *Xcr1*-DTR, B6. *Clec4c*-DTR, or B6.WT TCD-BM from donors as per (A) on day 0 followed by 2 × 10^6 B6.WT CD4$^+$ T-cells on day 3. DT treatment is as shown in (A). GVHD survival (*n* = 12 – 13 per T-replete group from two independent experiments). (**C-D**) Lethally irradiated B6D2F1 (allogeneic) and B6.WT (syngeneic) mice were transplanted with 10 × 10^6 BM from B6. *Xcr1*-venus$^{+/-}$ mice with 2 × 10^6 T-cell from

B6.WT mice (*n* = 4 per group from one experiment). (**C**) Representative images of the colon and mLN from B6D2F1 recipients stained with XCR1-driven venus (XCR1$^+$ cDC: green), Siglec H (pDC: red), CD4 (CD4$^+$ T: yellow), CD8 (CD8$^+$ T: purple), CD45 (hematopoietic cells: white) and DAPI (nuclear cells: blue). Scales: 100 μm. (**D**) CD4$^+$ or CD8$^+$ cell counts within 15 μm distance from XCR1$^+$ cDC or SiglecH$^+$ pDC per 10,000 nuclei. Statistical analysis by log-rank test for survival data and one-way ANOVA on clinical score at the individual time point (A-B), Mann-Whitney U test (D) dictated by normality testing of the data (mean ± SEM). **P* <0.05, ***P* <0.01, ****P* <0.001.

Figure 3. XCR1$^+$ cDC control donor T-cell expansion in the gut.

Lethally irradiated BALB/c mice were transplanted with 10 × 10^6 TCD-BM from DT-treated B6.*Xcr1*-DTR, B6.WT or B6.MHCII-KO mice and 0.2 × 10^6 T-cells from B6.WT mice. 3 ×10^6 TEa^{luc+} cells were transferred on day 12. Recipient mice were treated with DT on alternate days from day 8 to analysis. (**A**) Experimental scheme (**B, C**) Bioluminescence imaging for whole body (**B**) and organs (**C**) 5 days after TEa^{luc+} transfer. Fold change of total flux was calculated by comparing with the mean of WT BM group (*n* = 6 – 8 / group from two independent experiments). Data represents mean ± SEM. One-way ANOVA dictated by normality testing of the data (mean ± SEM). **P* <0.05, ***P* <0.01, ****P* <0.001.

Figure 4. XCR1$^+$ cDC control gut homing and Th1 differentiation.

Transplantation, day 12 TEa cell transfer, and DT injections were performed as in Figure 3 with analysis 5 days after TEa transfer. (**A**) Representative FACS plots of cDC subsets and the percentage of CD103$^+$CD11b^{neg} cDC from the mLN, spleen, colonic epithelium and lamina propria (*n* = 4 / group from one experiment). (**B**) Frequency of YAe$^+$ cDCs in the mLN. (**C**) YAe expression

by DC subsets in the mLN. The frequency of YAe^{+} cells in each subset shown by fold change, calculated by comparing to the mean of WT CD103^{+}CD11b^{neg} cDCs. (**D**) Representative histogram and frequency of integrin α4β7 expression on TEa cells. (**E**) Absolute numbers of α4β7^{+} TEa cells in mLN. (**F**) Frequencies of IFNγ, IL-17A and TNF producing TEa cells (*n* = 8 / group from two independent experiments). Statistical analysis by Mann-Whitney U test (A, E), unpaired t-test with Welch's correction (B,D,F), and ANOVA tests for multiple comparison (C), dictated by normality testing of the data (mean ± SEM). *P <0.05, **P <0.01, ***P <0.001, ****P <0.0001. N/A: not applicable since the relevant population was deleted.

Figure 5. WIN 18,446 administration inhibits donor CD11b^{+} cDC and alloantigen-specific Th17 expansion.

Recipients were treated with WIN18,446 or control diet from day -7 to day 70 (**A**) or analysis (**B – G**). (**A**) BALB/c mice were transplanted with B6 TCD BM and TEa as in Figure 2A. Kaplan-Meier survival curves compared by log-rank test (*n* = 6 – 16 / group from two independent experiments). (**B**) BALB/c mice were transplanted with 10 x10^{6} TCD-BM from B6.CD11c-luciferase (CD11c-GCDL) mice and 0.2 x 10^{6} T-cells from B6.WT mice. Representative bioluminescence images and radiance quantified on day 13 (*n* = 10 / group from two independent experiments. (**C-G**) Transplantation with B6.WT BM (**C-F**) or B6.*Xcr1*-DTR BM (**G**) and B6.WT T-cells on day 0, then day 12 TEa^{luc+} transfer as in Figure 3. (**C**) Representative bioluminescence images and quantification of radiance (*n* = 10 / group from two independent experiments). (**D**) Frequency and absolute number of DC subsets (*n* = 14 / group from three independent experiments) from the mLN 3 days after TEa transfer. (**E**) Representative bioluminescence images and quantification of radiance (*n* = 10 / group from two independent experiments). (**F**) Frequency and absolute number of IFNγ and IL-17A producing TEa cells in the mLN (*n* = 10 / group from two independent experiments). (**G**) BMT recipients receiving WIN18,446 diet or control diet were treated with DT

or vehicle from day 8 – 16 with analysis 5 days after TEa transfer. Absolute numbers of donor cDC and subsets, total and cytokine (IFNγ and IL-17A)-producing TEa T-cells in the mLN (*n* = 9 – 10 / group from two individual experiments). Statistical analysis by Mann-Whitney U test (B, D, IFNγ^{+} TEa count in F, IL-17A^{+} TEa count in G), unpaired t-test with Welch's correction (C, E, F, G) dictated by normality testing of the data (mean ± SEM). **P* <0.05, ***P* <0.01, ****P* <0.001.

Figure 6. WIN18,446 suppresses ALDH1A1 and cDC dependent alloreactive T-cell expansion and Th17 differentiation.

Lethally irradiated BALB/c mice were transplanted with 10 x 10^{6} BM from B6.WT or B6. Aldh1A1$^{-/-}$ mice with 0.2 x10^{6} B6.WT T-cells with WIN18,446 or control diet treatment (from day -7 to day 17). TEa^{luc+} cells were transferred on day 12. Bioluminescence imaging of whole body and organs, flow cytometric analysis of the mLN on day 12 + 5 (*n* = 7 – 8 / group from two individual experiments). (**A**) Representative bioluminescence images and quantification of radiance of whole body, gut and mLN. (**B**) Frequency and absolute number of TEa, IFN-γ and IL-17A expressing cells in the mLN. (**C**) Frequency and absolute number of integrin α4β7 expressing TEa cells. Statistical analysis by Kruskal-Wallis test (A-B), %IL17A with two-way ANOVA (B), dictated by normality testing of the data (mean ± SEM). **P* <0.05, ***P* <0.01, ****P* <0.001.

Figure 1

Fig.1

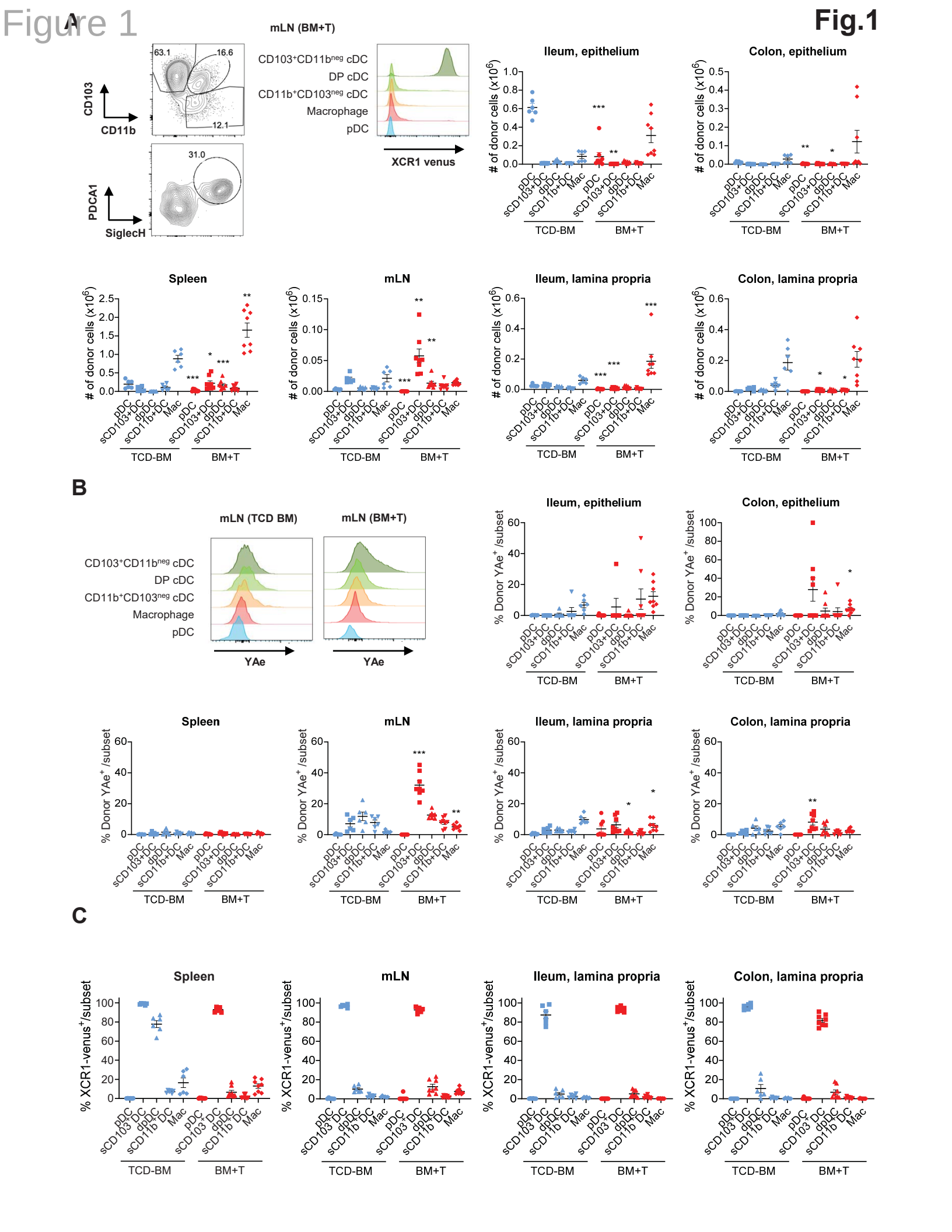
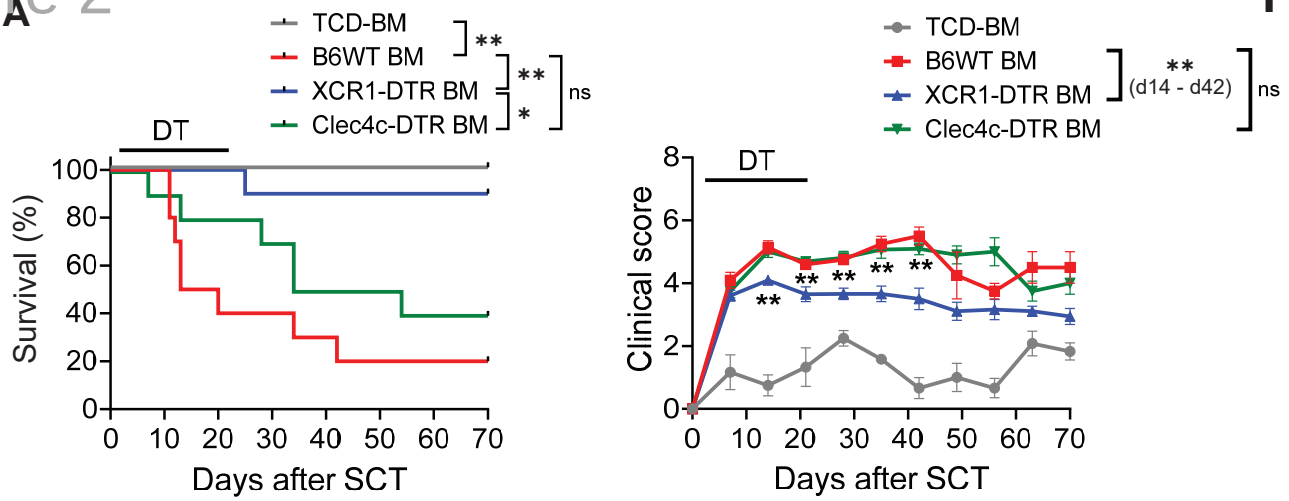


Figure 2

Fig.2

A



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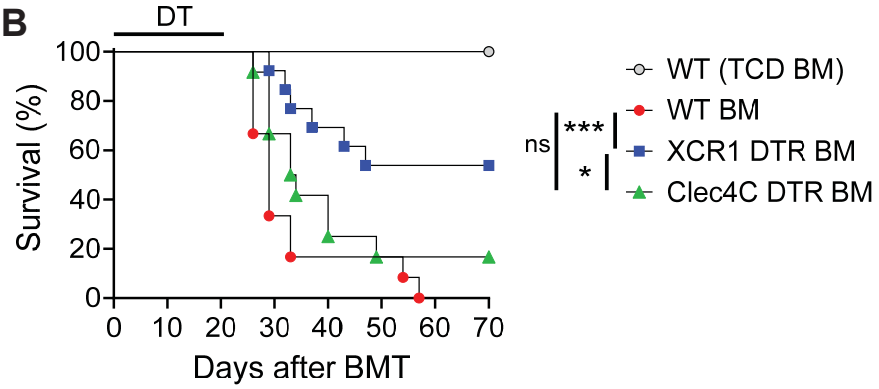

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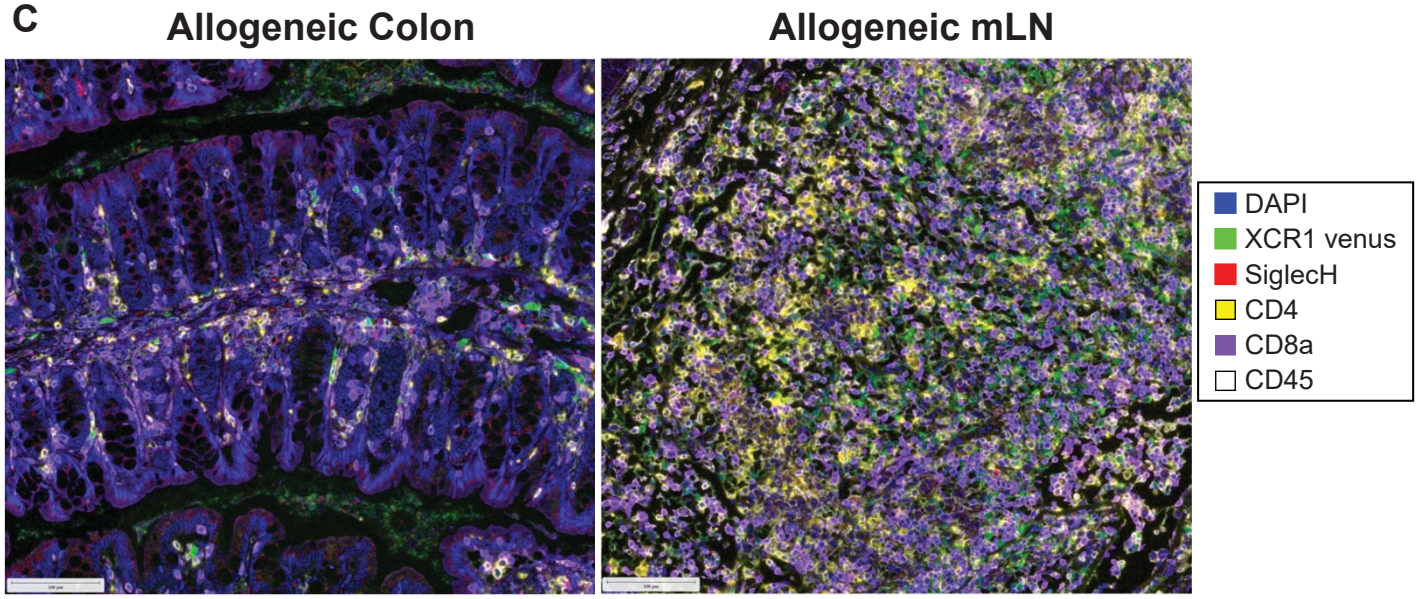

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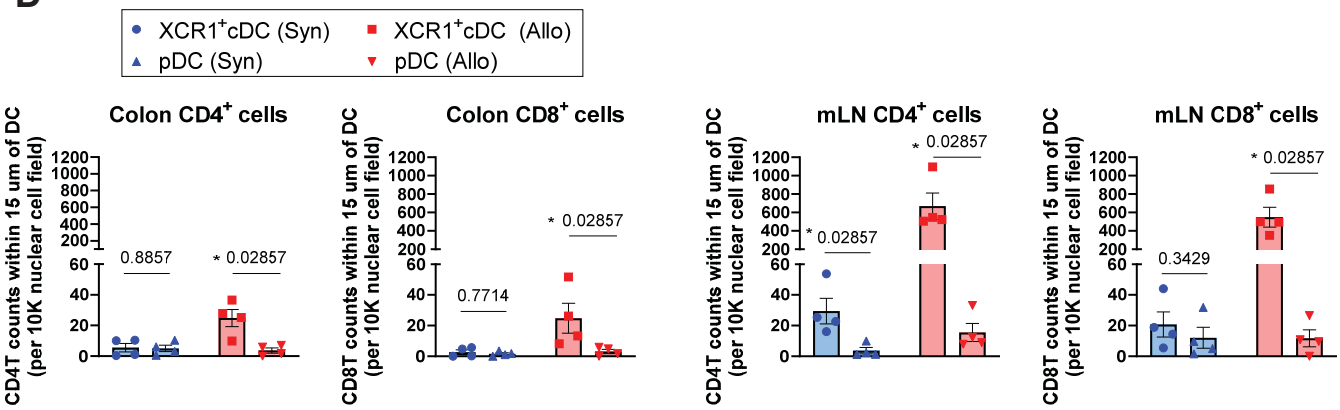


Figure 3

Fig.3

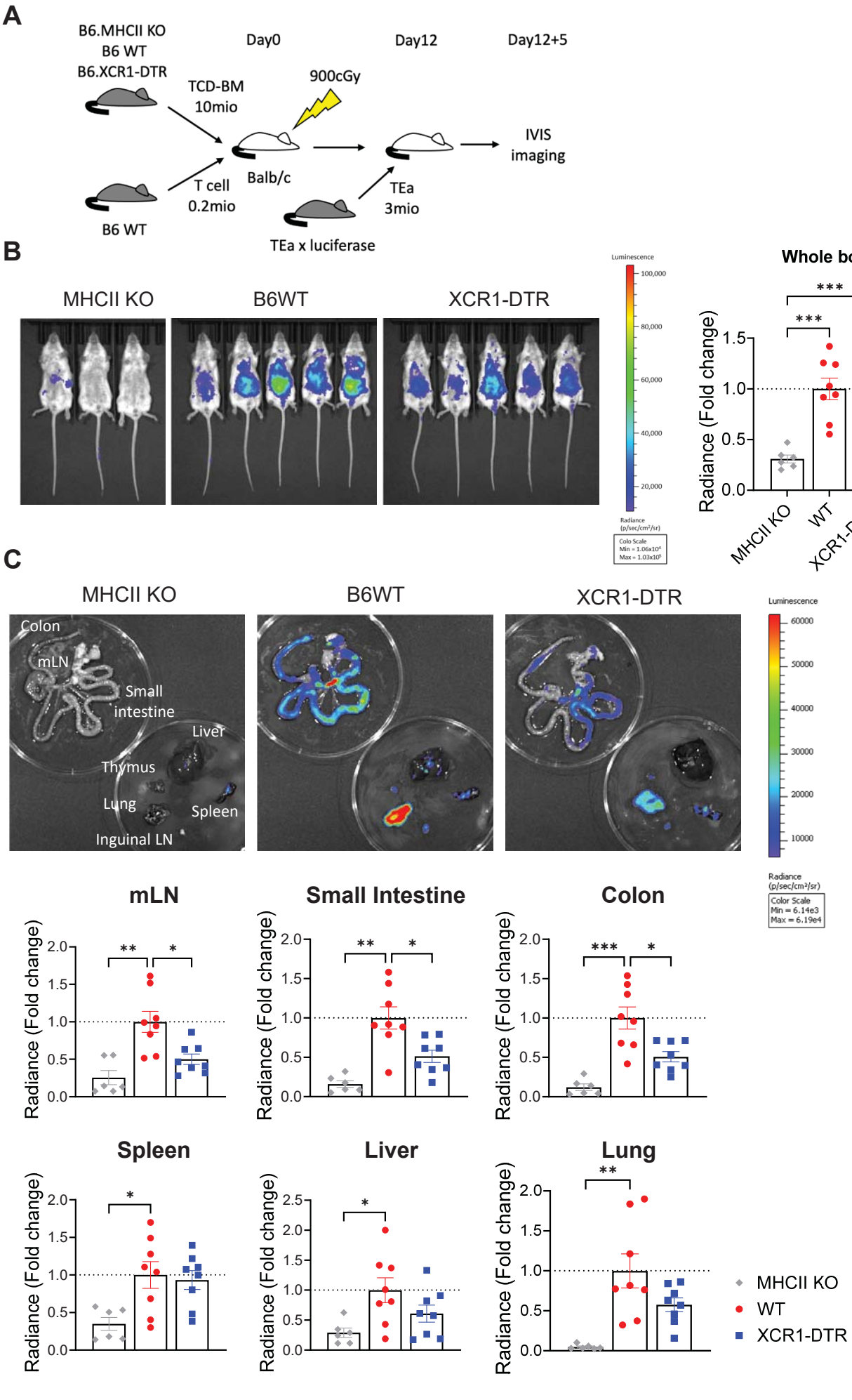


Figure 4

Fig.4

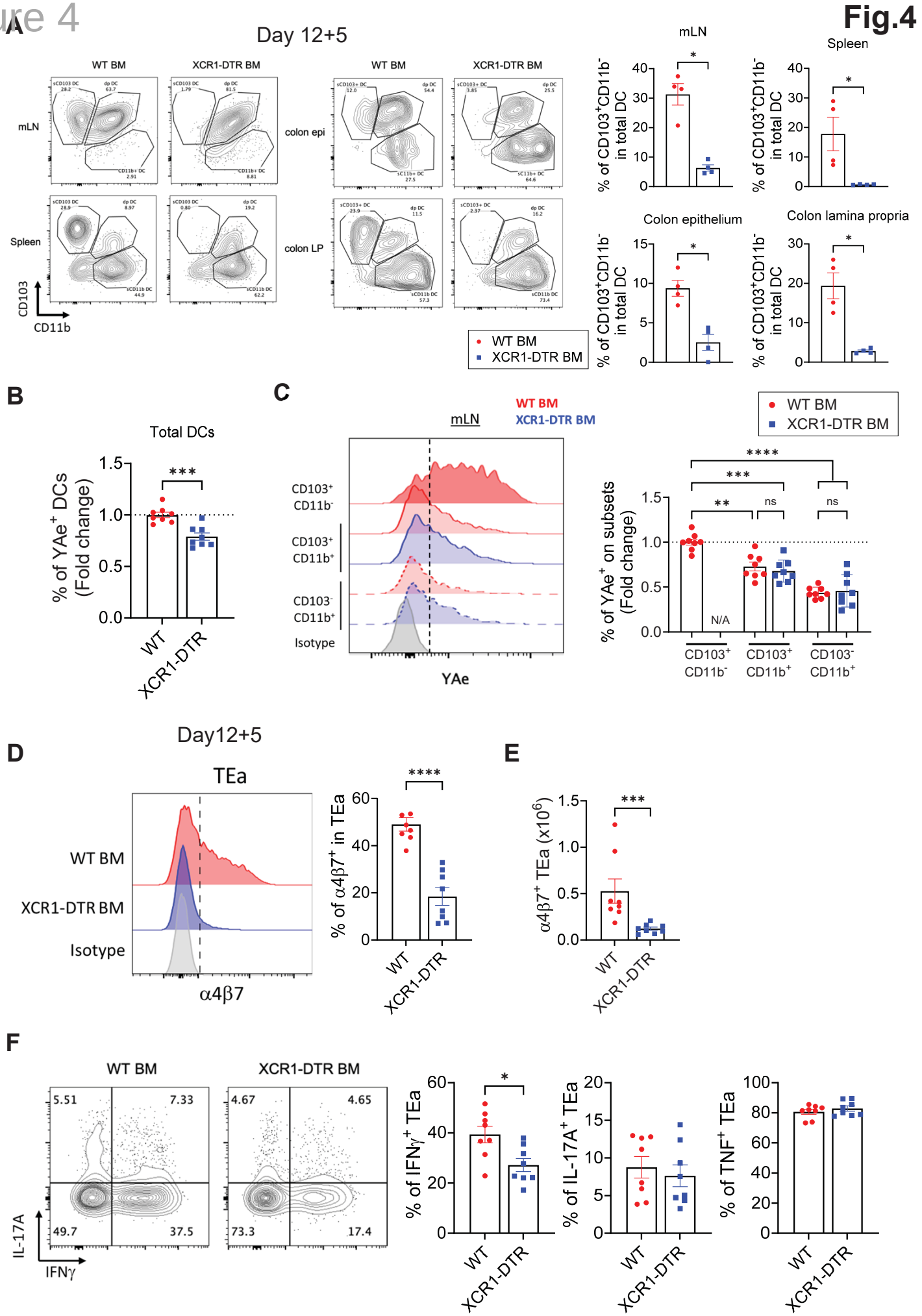


Figure 5

Fig.5

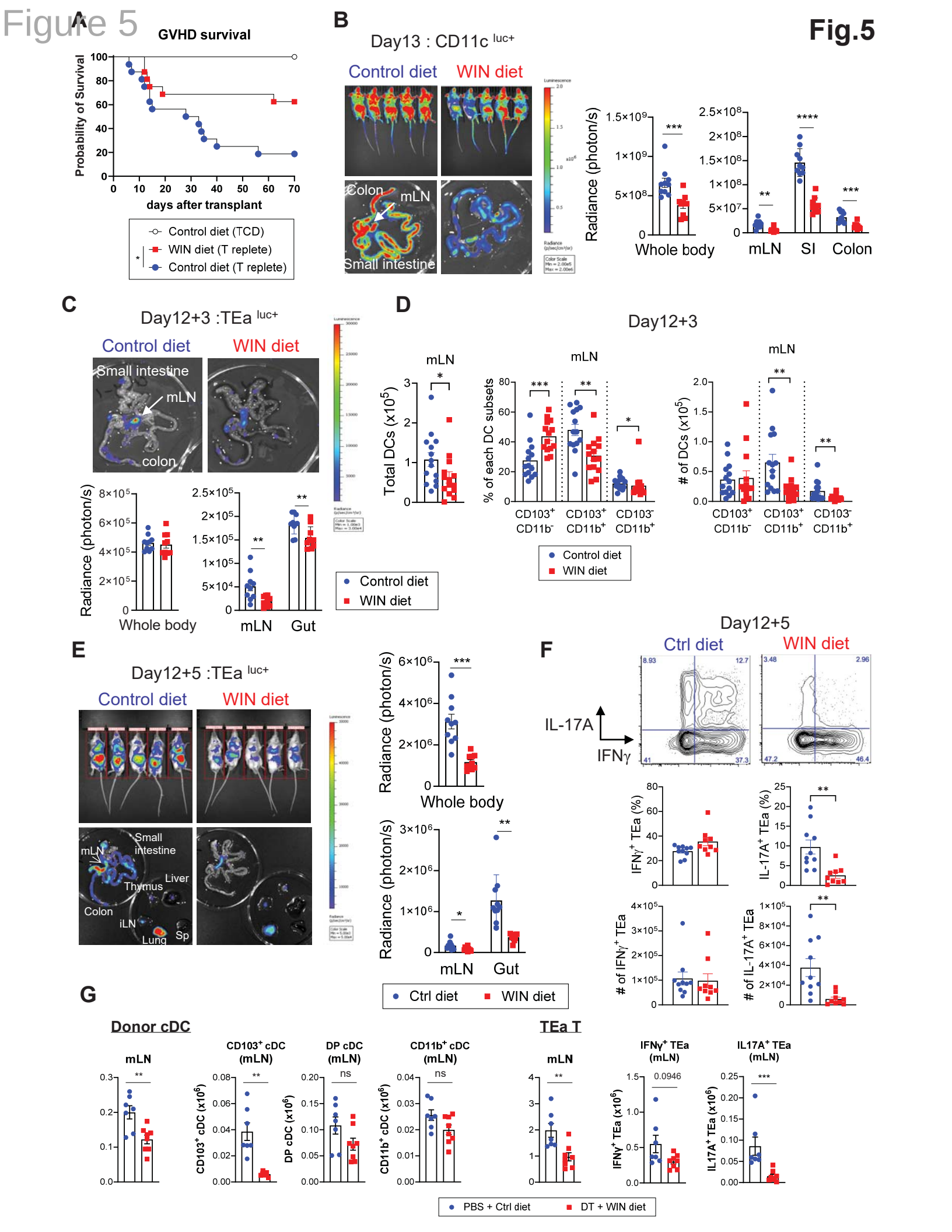
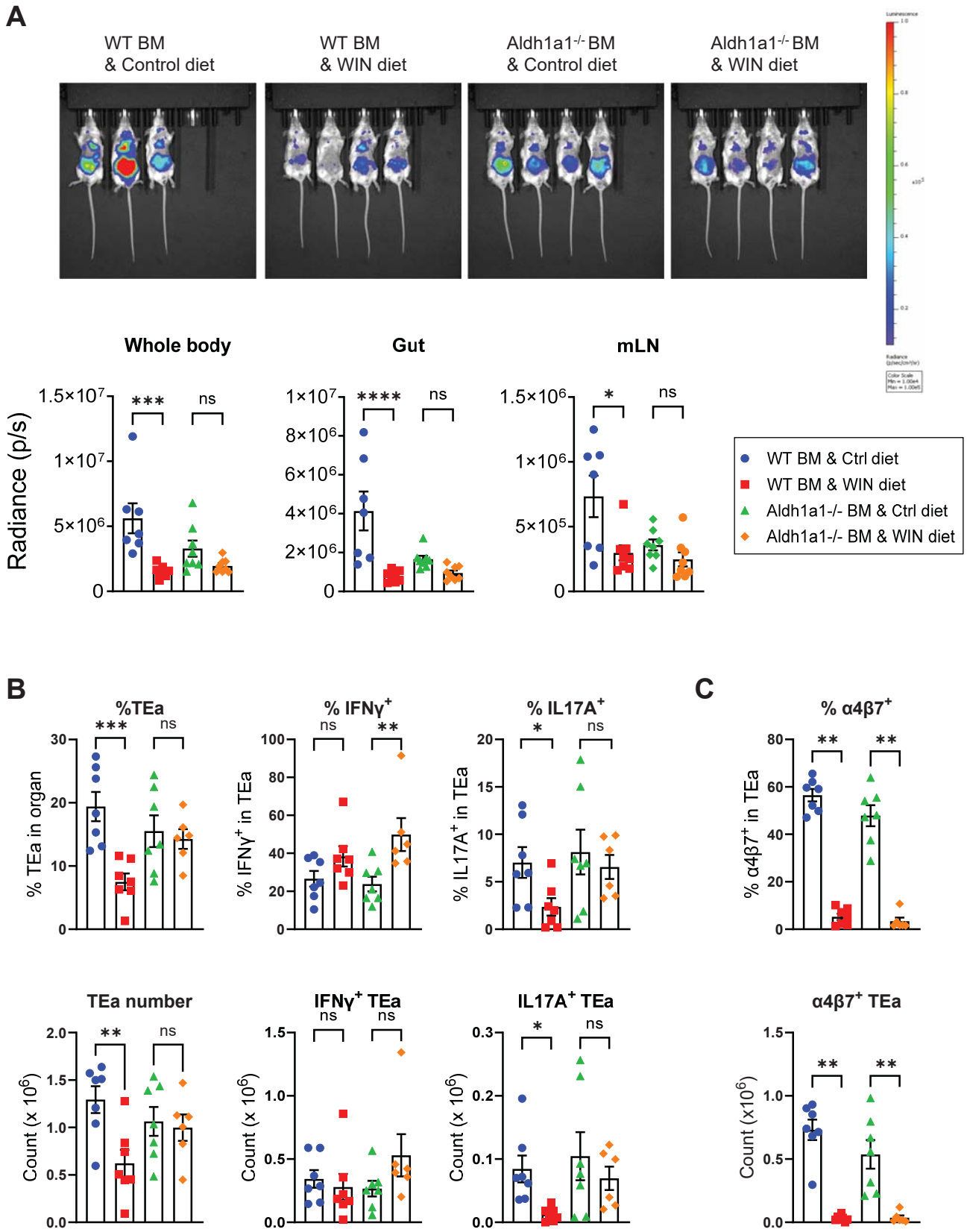


Figure 6

Fig.6



The Role of XCR1⁺ and CD11b⁺ Dendritic Cells in Gastrointestinal Graft-Versus-Host Disease (GVHD)

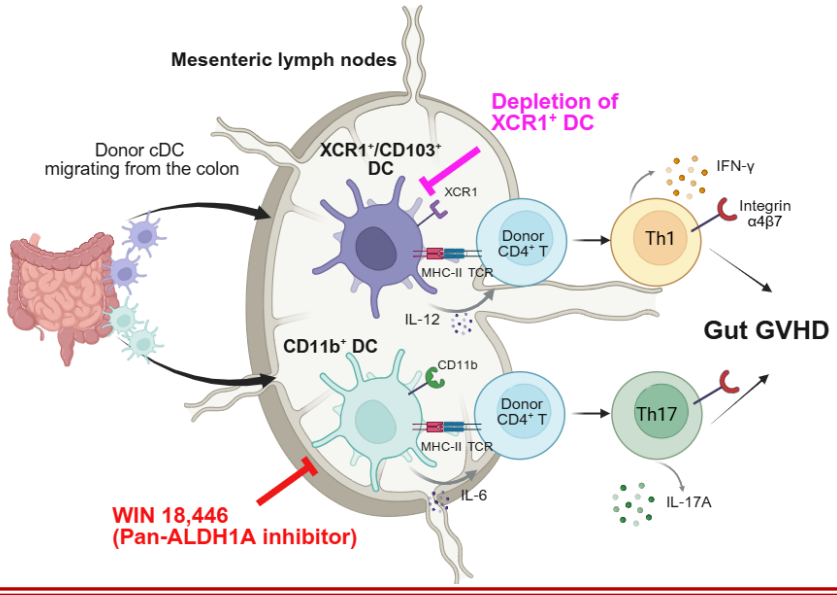
Context of Research

Although donor dendritic cells (DC) in the gut are known to amplify acute GVHD after allogeneic stem cell transplantation, the specific subsets and mechanisms involved are unclear.

Aim of This Study

The aim of this study was to investigate the role of various donor dendritic cell subsets in order to develop therapeutic strategies for preventing gut GVHD.

Findings



Conclusions: During gut GVHD, donor-derived XCR1⁺ conventional dendritic cells present high amounts of alloantigen, inducing Th1 differentiation and imprinting gut-homing integrin expression. Donor CD11b⁺ conventional dendritic cells induce Th17 differentiation, representing a second pathogenic axis.

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blood Visual Abstract